Environmental fluctuation limits gene regulation unless it can be tracked

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Abstract

Environmental variability is widely expected to favour more elaborate gene regulation, but models disagree even about the direction of the effect, and many of them measure regulatory complexity by counting network connections. We separated fluctuation from the information available for tracking it in R-Aevol, an individual-based model in which regulation arises from transcription factors encoded in a DNA-like sequence. We compared frozen and fluctuating environments and, under fluctuation, an informative cue, no cue, and a yoked cue that supplies the same signal proteins at the same frequencies but independently of the environment. Four predictions, committed to version control before the simulations were run and tested at three switching rates, were all supported. With an informative cue, fluctuation increased total regulatory strength 1.8- to 2.7-fold relative to a frozen environment, and informative cues produced 2.3 to 2.9 times the regulatory strength of yoked cues but no more genes. Without a cue, fluctuating populations had an eighth to a quarter of the frozen regulatory strength and fewer genes, whether an environment lasted half a lifetime or about 250 generations. About half the strength of fluctuating informative-cue populations was the signal proteins' own links, as a second pre-registered experiment confirmed. The genome-encoded network grew for tens of thousands of generations in frozen environments but hardly grew under fluctuation without a cue or with a yoked cue, and with an informative cue it became at least as strong as in a frozen environment with the same signals. In a third pre-registered experiment, frozen networks moved into cue-free fluctuation lost 77% of their strength, and fluctuating networks moved into a frozen environment grew. Fluctuation therefore limits gene regulation unless the environment can be tracked, and a usable cue is wired into the network. Link counts missed this, because frozen populations carried many weak links from the signal proteins.

Author summary

Organisms in changing environments often have complex gene regulation, and it is easy to assume that environmental change is what builds it. We checked this assumption in a computer model of evolving genomes. Populations evolved in environments that were either constant or switching between two states, and we controlled what the organisms could know about the switching. Some populations received a signal that told them which state they were in. Others received the same signals arranged so that they told them nothing, and a third group received no signal. In a constant environment the regulatory network grew over tens of thousands of generations. Switching without a meaningful signal kept it small, whether the

environment switched within a lifetime or only about every 250 generations, and populations moved from a constant into a switching environment lost most of their network. The uninformative signal did not change this, although it was physically identical to the meaningful one. With a meaningful signal, the network grew at least as strong as in a constant environment, and the signal itself was wired into it. Counting regulatory connections missed much of this. Our main predictions were written down and committed before the simulations were run.

Introduction

Organisms that live in variable environments are often assumed to need more elaborate gene regulation, and therefore to evolve it. The argument is simple. Regulation lets one genome produce different phenotypes in different conditions, and in a constant environment there is little for it to do. Comparative genomics offers some support. In bacteria the number of transcription factors grows faster than the number of genes [1], roughly as its square [2]. Free-living species carry more transcription factors than their genome size predicts [3], and intracellular parasites and other species of stable niches have simpler signalling systems than species from the open environment [4]. Simulations with R-Aevol, however, reproduce such scaling laws, trace them to differences in mutation and rearrangement rates between lineages and find that lifestyle does not explain them [5], so the comparative evidence alone does not show that variable environments select for regulation.

At the level of the phenotype, theory specifies when variability favours a response. A plastic response within a lifetime pays when the environment changes on the timescale of an individual's life and a reliable cue indicates the coming state [6–9]. Without such a cue, fluctuation favours a fixed generalist phenotype, bet-hedging or genetic tracking [8,9], and responding to a cue that predicts the environment poorly amplifies the mismatch between phenotype and environment [10]. A recent model of a gene regulatory network shows the same pattern for the evolution of plasticity. Plasticity evolved mainly under fast, erratic change, especially when cues were reliable, and evolvability under intermediate variability with less reliable cues [11].

Much less is known about the consequences for regulatory architecture, by which we mean the number of genes and the strength with which they regulate one another, and on this point models disagree. Tsuda and Kawata found that fluctuating selection promoted complex regulatory networks, mostly through the fixation of beneficial gene duplications, when the optimum moved in a random walk. When it alternated between two positions the effect was slight and limited to large, infrequent shifts [12]. Their networks received an external signal that told them when to express the phenotype but carried no information about where the optimum was. Bentkowski and colleagues reported larger genomes in fluctuating environments [13]. In a model in which network size was fixed, however, smaller networks let populations persist longer when the environment changed fast [14]. In another model with fixed numbers of genes and regulators, environmental change made regulatory complexity, measured as how evenly the regulators share control of each gene, approach the same level faster, and populations evolving without selection reached that level too [15]. Studies with the Aevol platform, on the other hand, find genome size limited by selection for mutational robustness and reduced by large population size and high mutation rates [16,17]. Work on modularity shows that its evolution depends on sparseness and on how the variation is structured [18,19].

Differences of method make these results hard to compare. Fluctuation and information are usually confounded. An environment that changes more often also contains a different amount of predictable structure, and adding a cue changes what an organism experiences as well as what it can know. A comparison of networks with and without a cue input [20,21] changes the two together. Studies of animal learning handle this problem with a yoked control, in which a second subject receives the same stimuli as the first but without the contingency between stimulus and outcome. Church showed that pairing individual subjects in this way can bias the comparison [22]. In our design each yoked arm receives the same stimulus statistics as the informative arm without being matched to particular informative-arm populations, so this bias does not arise. Varying the reliability of a cue achieves the same separation only if it includes a cue that is present at the usual frequencies but carries no information. Jorritsma and van den Berg included a cue of zero reliability, drawn uniformly and independently of the environment [11], but its values were not matched to those of their informative cues, and we know of no yoked control matched in stimulus statistics in studies of regulatory network evolution. Measurement is a further problem. Regulatory complexity is often quantified by counting genes or connections [12,18], but network size and connectivity can grow without selection for them [23], and a count does not reflect how strongly a connection acts. Weighted measures have been used [24], though seldom as the main outcome.

We addressed these issues in R-Aevol, a model of genome evolution in which regulation arises from transcription factors encoded in a DNA-like sequence that bind upstream of genes [5,25]. We compared frozen and fluctuating environments, and under fluctuation an informative cue, no cue, and a yoked cue

that presents the same signals at the same frequencies and with the same per-age rates of target and signal change as the informative one, but independently of the environment. Connectivity was measured as a count and as a strength. The study began with a different model, EvoSim, in which the genome is a regulatory network of evolvable size (S1, S4 and S6 Text). An exploratory experiment in R-Aevol contradicted one conclusion we had drawn from EvoSim, that fluctuation enlarges genomes. It reproduced another, that a cue's information does not change genome size, and showed in addition that information changes regulation. We then stated four predictions, committed them before running new simulations, and tested them at three switching rates and two population sizes. A further pre-registered experiment varied environmental grain, one of the differences between our design and that of Tsuda and Kawata. Finally, an exploratory analysis separated the links of the signal proteins from those of the genome-encoded network, a second pre-registered experiment tested the resulting pattern with new seeds, and a third moved populations between frozen and fluctuating environments.

Without information about the environment, fluctuation kept the regulatory network from growing as it did in a frozen environment. Without a cue this did not detectably depend on whether the environment changed within a lifetime or about every 250 generations, so environmental grain does not explain the difference from Tsuda and Kawata's results. A cue that carried no information did not change this. Moving populations between environments showed that fluctuation also removes networks that already exist and that networks grow again once it stops. With an informative cue, total regulatory strength rose, but about half of it came from the signal proteins' own links, and the genome-encoded network was at least as strong as in a frozen environment with the same signals. The confirmation experiment supported all four of its predictions about the genome-encoded network. Information did not increase genome size, and fluctuation reduced it. Connection counts did not track connection strength, so a count alone can misrepresent the effect of fluctuation on a network.

Results

Model and experimental design

R-Aevol is the version of the Aevol platform that includes gene regulation [5, 25, 26]. Genomes are circular sequences in which promoters, genes and binding sites are identified by motif search. Translated proteins contribute to a phenotype, a function that is compared with a target function, and can also bind upstream of promoters to activate or repress the transcription of other genes. Protein concentrations are integrated over the 20 ages of an individual's life, and fitness is evaluated against the target at each age. Environmental signals are signal proteins that are present at a fixed concentration during the ages in which their environment is active, so organisms can evolve to respond to them.

The design (Fig 1) crosses environment (frozen or fluctuating) with cue (informative, absent or yoked). In R-Aevol the fluctuating environment switches between two target functions, A and B. B is the mirror image of A and has the same geometric area, so neither target is harder to match. In the frozen environment the switching mechanism is kept but the two targets are identical. In the informative arm each target is paired with its own signal protein. The yoked arm has four environments, one for each combination of target and signal, so the two signals occur equally often but say nothing about the target. At each switch R-Aevol moves to one of the other environments at random. At a given switching probability the four-environment arm thus changes its target less often than the two-environment arm, and we multiplied the yoked arm's switching probability by 1.5 so that the two arms have the same per-age rates of target change and of signal change (Materials and methods). We write q for the switching probability per age.

The effect of fluctuation on regulation depends on the cue

Earlier exploratory work with EvoSim had suggested that fluctuation increases genome size and connectivity and that the information content of a cue makes no difference to genome size (S4 Text). R-Aevol reproduced the second conclusion but not the first. In an exploratory experiment (six arms, 12 seeds each, 30,000 generations), fluctuation reduced genome size under every cue condition, and its effect

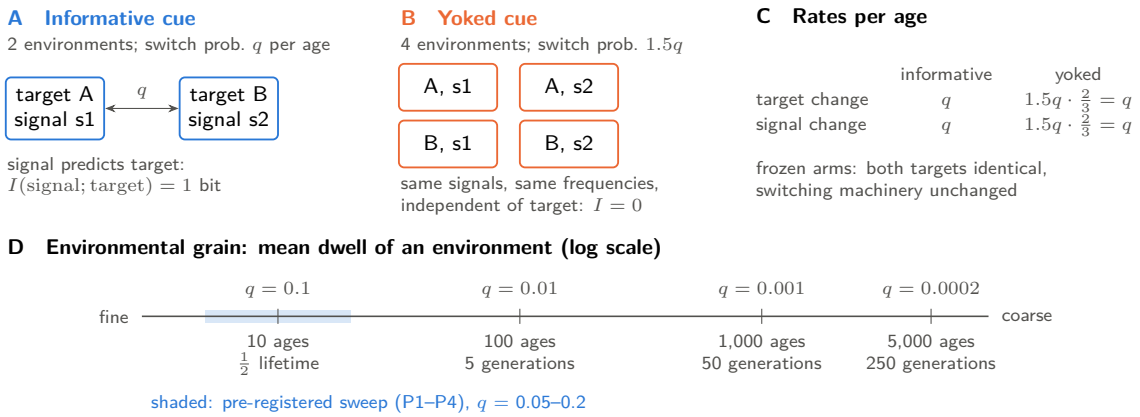


Fig 1. Design. (A) The informative arm in R-Aevol has two environments, each pairing a target with its own signal protein, so the signal predicts the target. (B) The yoked arm has four environments pairing each target with each signal, so the same signals occur at the same frequencies but carry no information about the target. (C) Because a switch moves to a different environment chosen at random, the yoked arm switches with probability $1.5q$, which gives it the same per-age rates of target change and signal change as the informative arm. Frozen arms use two identical targets. (D) Environmental grain as the mean dwell time of an environment, set by the switching probability q over the 20 ages of a lifetime. Ticks mark the four switching probabilities of the grain experiment (G1–G4); shading marks the range of the pre-registered sweep (P1–P4).

on regulation depended on the cue (Fig 2). With an informative cue, total regulatory strength was 12.7 in fluctuating populations and 4.7 in frozen populations given the same signals. Fluctuating populations without a cue had the lowest regulatory strength of the six arms (1.4).

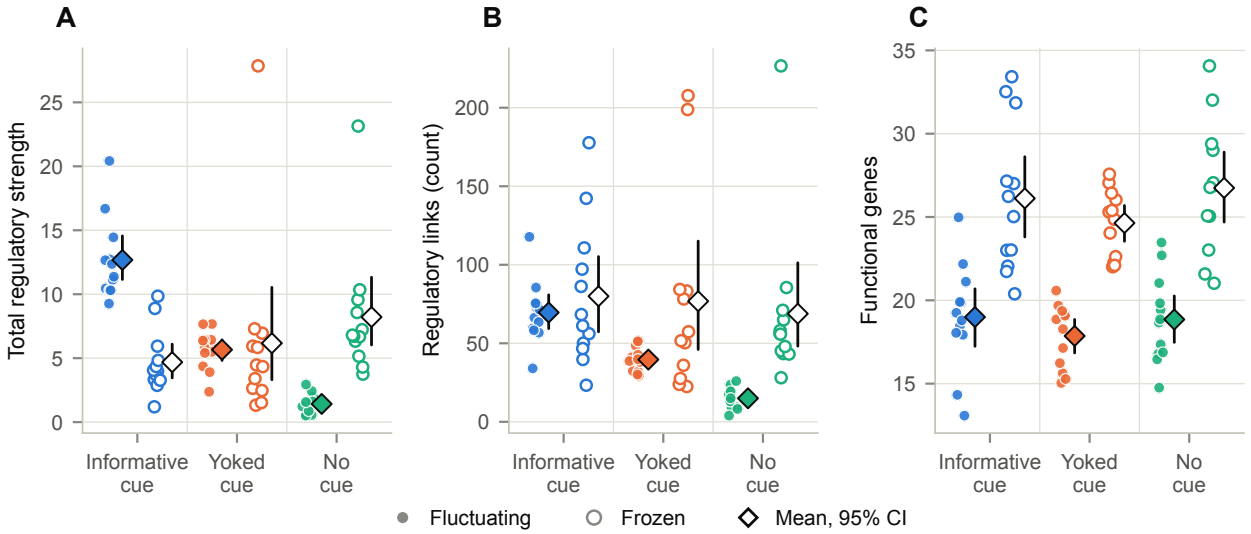


Fig 2. R-Aevol, six arms (exploratory). Total regulatory strength (A), number of regulatory links (B) and number of functional genes (C) for each combination of environment (filled, fluctuating; hollow, frozen) and cue (colour), one point per population; means with bootstrap 95% confidence intervals. $q = 0.10$, population 1024, 12 populations per arm.

Since this interpretation came from inspecting the data, we tested it with new seeds. Four predictions, with their measures, statistical tests and decision rules, were committed to version control before any of the new simulations were run (Table 1; Materials and methods). The switching probability took three values

over a fourfold range ($q = 0.05$, 0.10 and 0.20) at a population size of 1024, with 12 new seeds per condition, and the main contrasts were repeated at a population size of 512. For the two predictions that depend on cue use we also specified a manipulation check. At a given rate, the informative cue had to reduce metabolic error relative to the yoked cue, showing that the organisms used it. The check was met at $q = 0.10$ and $q = 0.20$ ($p < 10^{-4}$) but not at $q = 0.05$ ($p = 0.088$), so under the pre-registered rule these two predictions were evaluated at the two faster rates.

Table 1. Pre-registered predictions and verdicts.

Id	Prediction	Verdict
P1	Fluctuation with an informative cue raises total regulatory strength over frozen	supported ($q = 0.10$, 0.20)
P2	An informative cue raises it over a yoked cue	supported ($q = 0.10$, 0.20)
P3	Fluctuation without a cue lowers it below frozen	supported (all rates)
P4	Fluctuation with an informative cue lowers the number of functional genes	supported (all rates)
G1, G3	Coarse-grained cue-free fluctuation gives more genes (G1) and stronger regulation (G3) than fine-grained	inconclusive
G2, G4	Coarse-grained cue-free fluctuation gives more genes (G2) and stronger regulation (G4) than frozen	refuted
C1	Without a cue, fluctuation weakens the genome-encoded network below frozen	supported ($q = 0.10$, 0.20)
C2	With a yoked cue, fluctuation weakens it below frozen with the same signals	supported ($q = 0.10$, 0.20)
C3	An informative cue keeps it stronger than a yoked cue	supported ($q = 0.10$, 0.20)
C4	With an informative cue it is not more than 25% weaker than frozen	supported ($q = 0.10$, 0.20)
W1	A frozen network moved into cue-free fluctuation is lost (or persists)	lost (ratio 0.21)
W2	A fluctuating network moved into a frozen environment grows	grows

P1–P4 were supported if the predicted sign held at every evaluable rate and the Holm-adjusted $p < 0.05$ at two or more, refuted if the opposite sign was significant at any rate or no rate was Holm-significant, and inconclusive otherwise. P1 and P2 are evaluable only at rates where the informative cue measurably lowers metabolic error relative to the yoked cue. G1–G4 are single tests, supported if the predicted sign holds with Holm-adjusted $p < 0.05$, refuted if the opposite sign holds with $p < 0.05$, and inconclusive otherwise. C1–C3 were to be supported if the predicted sign held with Holm-adjusted $p < 0.05$ at $q = 0.10$ and at $q = 0.20$, and C4, a non-inferiority test, if the lower bound of the 95% bootstrap confidence interval of the ratio of means exceeded 0.75 at each rate. C3 and C4 were evaluable only where the informative cue lowered metabolic error, as it did at each rate. W1 and W2 compare switched with unswitched populations from the same source at the end of the experiment, with Holm correction across the two; W1 would have read as persistence if the difference was not significant and the lower bound of the 95% confidence interval of the ratio exceeded 0.75.

All four predictions were supported (Table 1, Fig 3). With an informative cue, fluctuation increased total regulatory strength relative to the frozen environment 1.8-fold at $q = 0.10$ and 2.7-fold at $q = 0.20$ (P1; Holm-adjusted $p = 0.005$ and $p \approx 10^{-4}$). Without a cue, fluctuation lowered regulatory strength about fivefold at all three rates (1.1 to 1.2 compared with 6.3; P3; Holm-adjusted $p \approx 10^{-4}$ at each rate). The informative-cue cells at the slowest rate, which were excluded from the verdict, showed the same direction (11.0 compared with 5.9). In one of those populations every regulatory connection was lost for the whole analysis window, which happened in no other population of the sweep.

The information carried by the cue produces the effect

The yoked control separates information from stimulation. Informative and yoked arms receive the same two signal proteins at the same concentration and frequency, with the same per-age rates of target change and signal change, and differ only in whether the signal predicts the target. Informative-cue populations evolved 2.3 times ($q = 0.10$) and 2.9 times ($q = 0.20$) the regulatory strength of yoked-cue populations (P2; Holm-adjusted $p \approx 10^{-4}$ in each case) and matched the target about twice as closely (metabolic error 0.029 against 0.058 at $q = 0.10$). They also carried more transcription factors at every rate (11.2 against 6.2 in the exploratory experiment and 10.6 against 7.4 at $q = 0.10$ in the sweep). In the frozen environment the

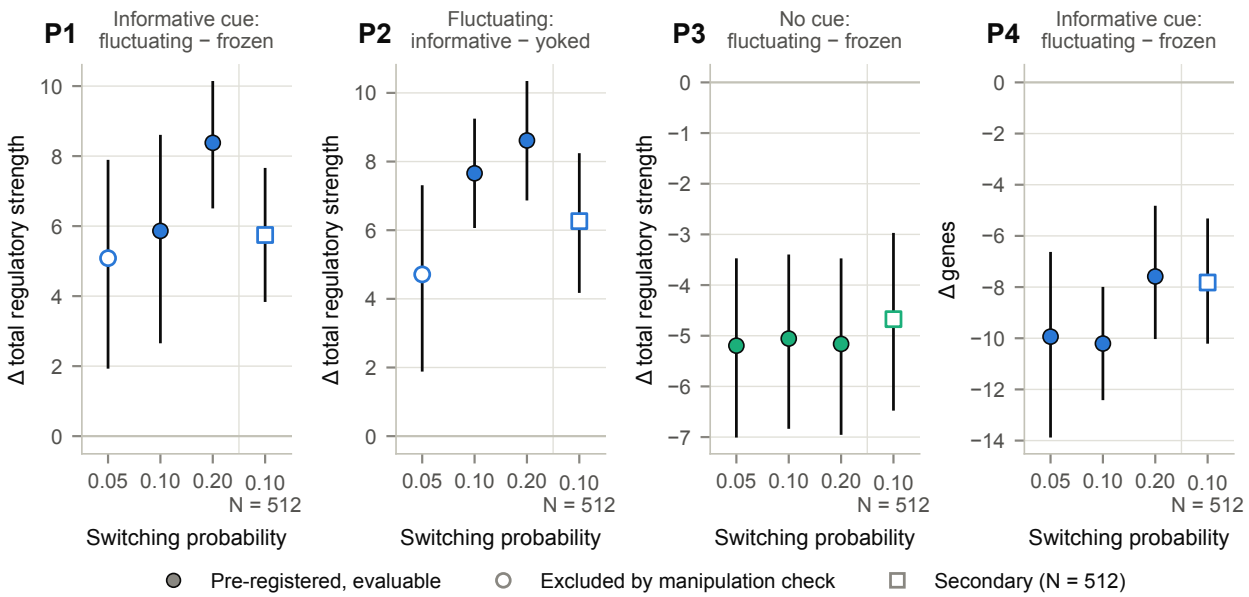


Fig 3. Pre-registered test. Differences in means with bootstrap 95% confidence intervals for each pre-registered contrast, at each switching probability (population 1024) and at population 512 (squares; secondary, no verdict). Open circles mark cells excluded from the verdict because the manipulation check failed at that rate. For P3 at population 512 the frozen baseline is the population-1024 frozen arm, the only one run. P1: effect of fluctuation on total regulatory strength with an informative cue. P2: informative minus yoked cue under fluctuation. P3: effect of fluctuation without a cue. P4: effect of fluctuation on the number of functional genes.

signal still switches but predicts nothing, because the target does not change. There the informative and yoked cues did not differ significantly in the pre-registered sweep (7.5 and 5.4, $p = 0.18$), and in the exploratory experiment the difference went the other way (4.7 and 6.2). The difference between cue arms appears only when the cue carries information about a changing environment.

The information did not increase genome size. Informative- and yoked-cue populations carried similar numbers of functional genes in the exploratory experiment (19.0 and 17.9, $p = 0.30$) and in the sweep (16.3 and 16.4, 18.1 and 19.9, and 18.0 and 20.8 at the three rates), where the only significant difference, at $q = 0.20$, favoured the yoked cue ($p = 0.01$, uncorrected). This agrees with the earlier EvoSim conclusion, which held even in a regime in which organisms read the cue. With reversible development, informative-cue populations matched the best-paying resource channel in 1.5 of 6 states, against 1.0 with a scrambled cue, yet had the same number of nodes (S4 Text). What information changed in R-Aevol was regulation, which has not been measured in that EvoSim regime.

Much of the gain is the cue's own wiring

R-Aevol treats the signal proteins like other proteins when it computes binding, so the regulatory strength and link counts above include the influences of the signal proteins. To separate them from the genome-encoded network, we continued every run from its saved population at generation 30,000 for 2,000 generations, with statistics that record the signal proteins' influences separately (Materials and methods). This analysis was not pre-registered. The continuations reproduced the regulatory strength of the analysis window (the correlation of cell means was 0.995 across 21 cells).

About half of the regulatory strength of fluctuating informative-cue populations came from the signal proteins (50 to 55%; Fig 4A). Their genome-encoded network was no stronger than in the frozen environment at $q = 0.05$ and 0.10 (5.4 against 4.8 and 5.9 against 5.4; $p = 0.61$ and 0.62) and 1.5 times stronger at $q = 0.20$ (6.8 against 4.4, $p = 0.007$). The gain in total strength tested by P1 is therefore

mainly the cue being wired into the network. The yoked cue differed. Its signal proteins also carried strong links, which made up 68 to 80% of the total, but the genome-encoded network was as weak as in populations without a cue (0.9 to 2.1 against 1.1 to 1.3; $p = 0.11$, 0.06 and 0.67), less than half its strength in any frozen arm of the sweep (4.4 to 6.4). On the genome-encoded network alone, the informative cue produced 2.5, 3.7 and 7.2 times the strength of the yoked cue ($p \leq 0.007$), larger ratios than for total strength, and the difference held at a population size of 512 and in the exploratory experiment. Without information, then, the genome-encoded network stays weak under fluctuation whether or not a signal is present, and with information it is at least as strong as in a frozen environment.

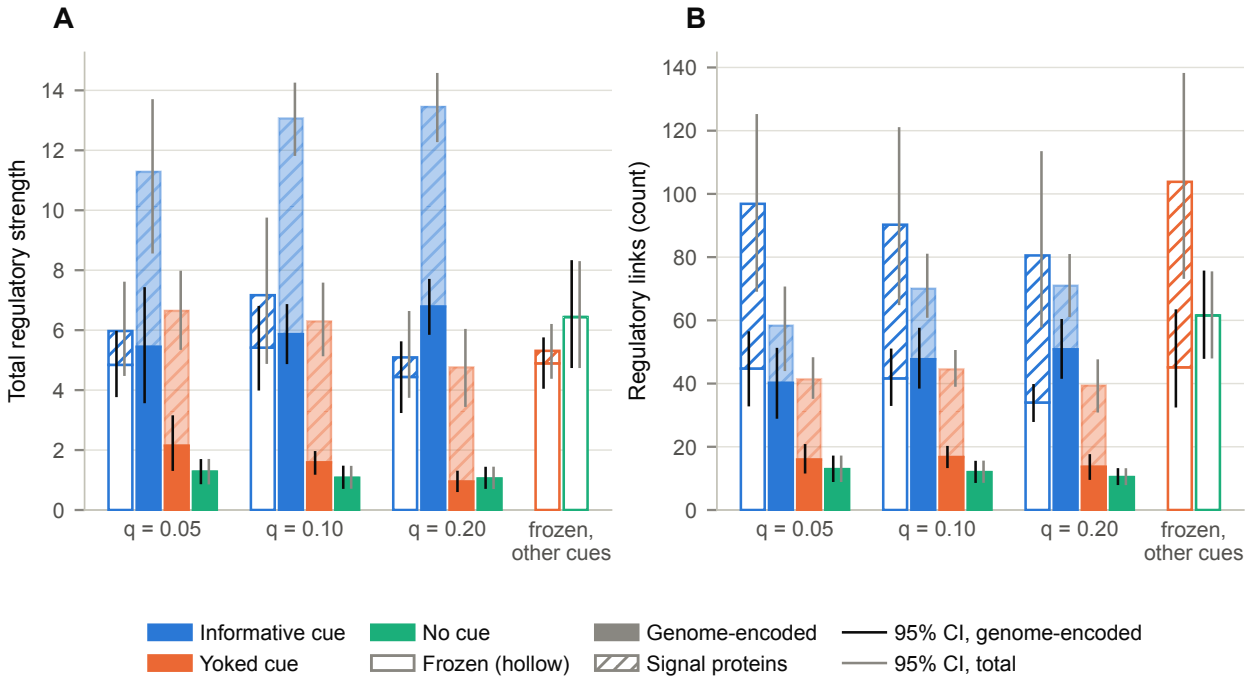


Fig 4. Signal-protein links and the genome-encoded network (exploratory). Continuations of every run from its saved population at generation 30,000 to 32,000, population 1024. (A) Total regulatory strength divided into the part exerted by the signal proteins (hatched) and by genome-encoded proteins (solid). (B) The same for the number of regulatory links. Colour gives the cue; frozen arms are hollow. Whiskers are bootstrap 95% confidence intervals of the genome-encoded part (black) and of the total (grey). The frozen yoked and frozen no-cue arms were run at $q = 0.10$; the frozen no-cue arm does not depend on q.

A pre-registered confirmation

Because the decomposition was exploratory, we tested its pattern in a second pre-registered experiment (Table 1; Materials and methods). The instrumented version of the platform recorded the signal proteins' influences from the first generation, with 12 new seeds in each of 11 cells at $q = 0.10$ and 0.20. The manipulation check held at each rate ($p < 10^{-4}$), and all four predictions about the genome-encoded network were supported (Table 2). Without a cue, the fluctuating network had about a sixth of its frozen strength (1.3 against 7.9 at each rate; C1). With a yoked cue it reached 1.8 and 1.2, against 4.0 and 5.6 in frozen populations with the same signals (C2). The informative cue made it 3.5 and 5.0 times as strong as the yoked cue (C3) and at least as strong as in frozen populations with the same signals (C4). In fact it was 1.60 and 1.53 times as strong there (95% confidence intervals 1.13 to 2.28 and 1.11 to 2.14). This increase over the frozen environment, significant at each rate as a secondary comparison ($p = 0.016$ and 0.029, uncorrected), had appeared only at $q = 0.20$ in the exploratory continuations. The signal proteins again contributed about half of the regulatory strength of fluctuating informative-cue populations (53%).

One observation qualifies the frozen reference. Frozen populations without signals had a stronger genome-encoded network (7.9) than frozen populations with signals (3.9 to 5.6), as in the exploratory analysis. How informative-cue populations compare with a frozen environment therefore depends on whether that environment has signals, and the pre-registered comparison used the one that does.

Table 2. Pre-registered confirmation: genome-encoded regulatory strength.

Prediction	q	A	B	Test	Verdict
C1	0.10	1.34	7.88	Holm $p = 6 \times 10^{-5}$	supported
	0.20	1.31	7.88	Holm $p = 6 \times 10^{-5}$	
C2	0.10	1.84	4.03	Holm $p = 0.011$	supported
	0.20	1.20	5.64	Holm $p = 6 \times 10^{-5}$	
C3	0.10	6.53	1.84	Holm $p = 2 \times 10^{-4}$	supported
	0.20	5.96	1.20	Holm $p = 9 \times 10^{-5}$	
C4	0.10	6.53	4.09	ratio 1.60, 95% CI 1.13–2.28	supported
	0.20	5.96	3.88	ratio 1.53, 95% CI 1.11–2.14	

Mean genome-encoded regulatory strength over generations 25,001–30,000, 12 populations per cell. In C1, A is fluctuating and B frozen, without a cue; in C2, fluctuating and frozen with a yoked cue; in C3, informative and yoked cue under fluctuation; in C4, fluctuating and frozen with an informative cue. C1 to C3 are two-sided permutation tests with Holm correction across their six tests; C4 is a non-inferiority test on the ratio of means with a margin of 0.75 and a bootstrap confidence interval.

Frozen networks grew, and networks under uninformative fluctuation did not

Every run began with a single gene, so the comparisons above do not show whether fluctuation removed regulation or kept it from being built. The statistics of every generation show the second (Fig 5). In frozen environments the genome-encoded network grew for tens of thousands of generations, in the confirmation from 3.0 in the first 2,000 generations to 7.9 in the analysis window. Without a cue, fluctuating networks stayed small in every experiment. Over all 120 cue-free fluctuating populations, regulatory strength averaged 1.4 in the first 2,000 generations, 1.7 in generations 2,001 to 10,000 and 1.4 in the analysis window. On a logarithmic scale, 55% of the final difference from the frozen environment came from frozen networks growing after generation 2,000, 44% was already present within the first 2,000 generations, and 1% came from change in the fluctuating networks. With a yoked cue the network grew slowly (from 1.1 to 1.8 at $q = 0.10$), and with an informative cue it grew from 2.3 to 6.5.

The smaller networks had fewer regulators, and their links were about as strong. In the analysis window of each of the ten cue-free cells, frozen networks were 3.7 to 8.0 times as strong as fluctuating ones. Genome-encoded proteins with at least one regulatory link were 2.0 to 3.7 times as numerous in frozen populations, which accounts for 46 to 63% of the difference on a logarithmic scale, and each had 1.5 to 2.2 times as many links (26 to 46%). The mean strength of a link differed less (frozen links 0.96 to 1.39 times as strong). Smaller genomes explain only part of this. Regulator proteins numbered 61 to 65% of the functional genes in frozen populations and 28 to 44% under cue-free fluctuation (Fig 5C). These analyses of trajectories and composition are exploratory.

Moving populations between environments

The trajectories do not show whether fluctuation would remove a network that already exists. We tested this in a third pre-registered experiment (Table 1; Materials and methods). At generation 30,000 the best genome of each cue-free population of the confirmation at $q = 0.10$ founded a new population, which evolved for 10,000 generations in the same environment or in the other one. Frozen networks moved into cue-free fluctuation lost 77% of their strength, half of that within 500 generations (Fig 6). After 10,000 generations they had 0.21 times the strength of networks that stayed frozen (1.8 against 8.5; 95% confidence interval of the ratio 0.17 to 0.26; Holm-adjusted $p = 2 \times 10^{-5}$), close to that of populations that had always fluctuated (1.3), with 9 regulator proteins instead of 18 and 20 functional genes instead of 27. In the other direction, fluctuating networks moved into a frozen environment grew 3.9-fold, to 5.2 against

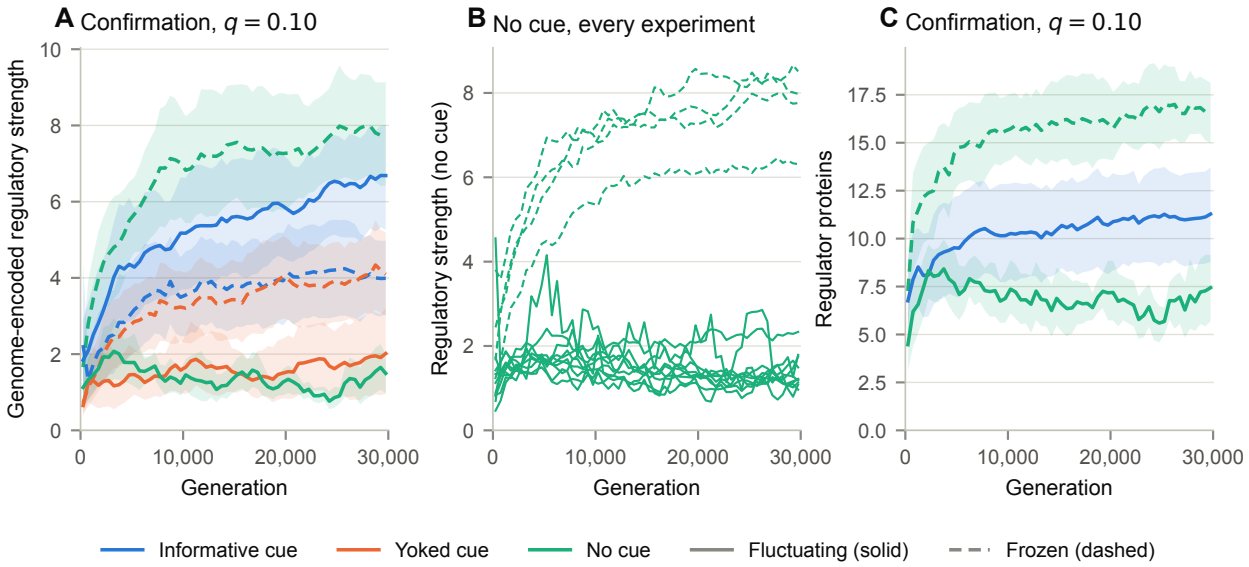


Fig 5. The genome-encoded network over 30,000 generations (exploratory). Means over consecutive blocks of 500 generations; frozen arms dashed, fluctuating arms solid, colour gives the cue. (A) Genome-encoded regulatory strength in every arm of the confirmation at $q = 0.10$ (12 populations each), with bootstrap 95% confidence bands. (B) Regulatory strength of every cue-free cell of the four R-Aevol experiments (exploratory, sweep, grain and confirmation), one line per cell of 12 populations; without signals, all links are genome-encoded. (C) Genome-encoded proteins with at least one regulatory link, confirmation, $q = 0.10$.

1.3 in populations that kept fluctuating (Holm-adjusted $p = 2 \times 10^{-5}$), with 16 regulator proteins instead of 7 and as many functional genes as frozen populations. They reached 62% of the frozen strength in 10,000 generations, less than frozen networks built from a single gene reached in the same time (7.0). Under the interpretation fixed in advance, uninformative fluctuation removes existing networks as well as keeping new ones from growing, and the limit is set by the current environment and does not depend on the history of the genome.

Fluctuation reduces genome size

With an informative cue, fluctuation reduced the number of functional genes at every switching rate, by 10, 10 and 8 genes (26.3, 28.3 and 25.6 in frozen populations compared with 16.3, 18.1 and 18.0 in fluctuating ones; P4; Holm-adjusted $p \leq 3.2 \times 10^{-4}$; Fig 3). The reduction was also present at a population size of 512 (25.3 compared with 17.5, $p < 10^{-4}$), and without a cue or with the yoked cue (S3 Table). Other models have reported the opposite, larger genomes under fluctuating selection [12, 13], and so did earlier EvoSim experiments (S4 Text). A possible explanation lies in how Aevol builds phenotypes. The phenotype is the sum of contributions from individual genes, so a constant target that can be matched precisely favours additional genes. A target that changes within a lifetime cannot be matched precisely, and the coarser phenotype it leaves needs fewer genes. Consistent with this, frozen populations reached a metabolic error of about 0.008 and fluctuating populations without a usable cue about 0.06. We have not tested this explanation directly, and two results do not support it. Coarse-grained populations matched their targets better than fine-grained ones (metabolic error 0.038 against 0.062) but did not carry more genes, and informative-cue populations matched theirs better than cue-free ones (0.029 against 0.063 at $q = 0.10$) with fewer genes (18.1 against 20.0).

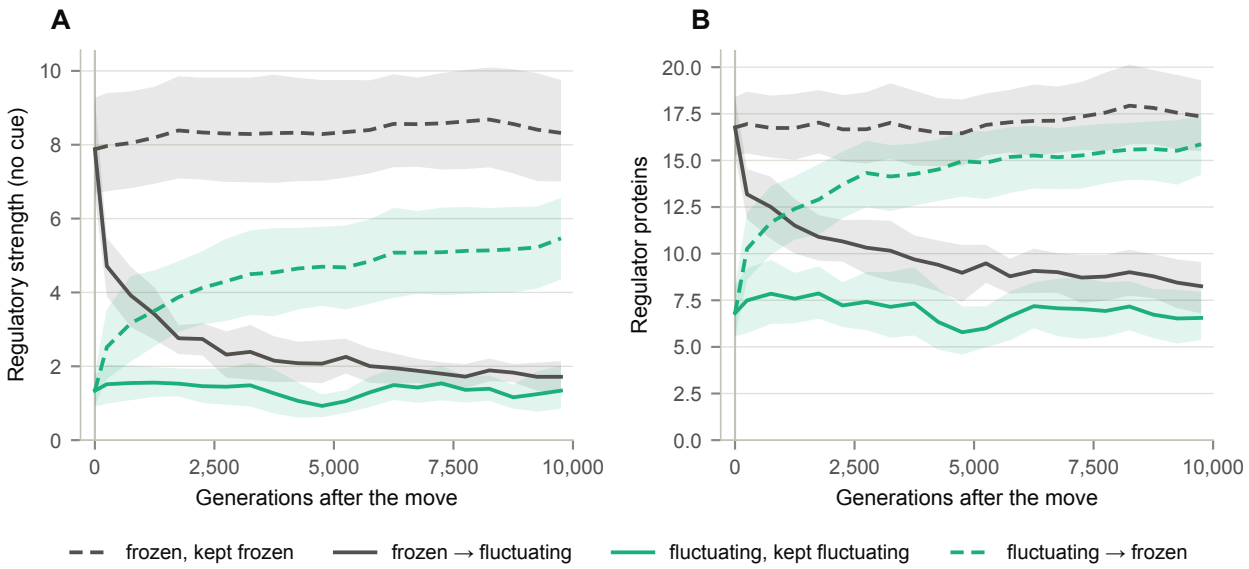


Fig 6. Populations moved between environments (pre-registered). The best genome of each cue-free confirmation population ($q = 0.10$) at generation 30,000 founded a population that evolved for 10,000 generations in the same or the other environment, 12 populations per cell. Generation 0 shows the source populations over generations 25,001–30,000; later points are means over blocks of 500 generations with bootstrap 95% confidence bands. Colour gives the origin (grey frozen, green fluctuating), line style the current environment (dashed frozen, solid fluctuating). (A) Regulatory strength; without signals, all links are genome-encoded. (B) Genome-encoded proteins with at least one regulatory link.

Connection counts do not track connection strength

Link counts and regulatory strength did not change together (Fig 4B). In the analysis window, fluctuating populations with an informative cue had no more regulatory links than the corresponding frozen populations (0.60, 0.78 and 0.83 times as many at the three rates; $p = 0.02$, 0.13 and 0.52), although their total regulatory strength was 1.8 to 2.7 times higher. The decomposition shows why. The frozen populations carried many weak links from the signal proteins (47 to 52 links with a combined strength of 0.7 to 1.8), whereas fluctuating informative-cue populations carried fewer and much stronger ones (18 to 22 links with a combined strength of 5.8 to 7.2). In the genome-encoded network, link number followed strength. The two rose together at $q = 0.20$ (51 against 34 links, $p = 0.008$), and neither differed significantly at the slower rates. In the confirmation the genome-encoded network was stronger at $q = 0.10$ and at $q = 0.20$, and it also had more links (51 against 32 and 50 against 30; $p = 0.04$ and 0.003, uncorrected). A study measuring connectivity as link number would have missed the wiring of the cue into the network, and its point estimates would have suggested the opposite. Without a cue, number and strength fell together to about a fifth of the frozen values. The comparisons of link number in the analysis window were pre-specified as secondary analyses without a formal verdict, and the decomposition and the link counts of the confirmation are exploratory.

Environmental grain

Tsuda and Kawata [12] found that fluctuation across generations, with no information about the position of the optimum, promotes regulatory complexity. The effect was strong when the optimum moved in a random walk, and slight when it alternated between two positions as in our design. Their fluctuation was also slow relative to a lifetime, whereas ours was fast. To test whether grain explains the difference, we varied it in R-Aevol without modifying the model. The environment is a single Markov chain over ages that continues from one generation to the next, so the switching probability sets how long an environment persists on average, from half a lifetime at $q = 0.1$ to about 250 generations at $q = 0.0002$ (Fig 1D). We

predicted that coarse-grained cue-free fluctuation would produce larger genomes and stronger regulation than fine-grained fluctuation (G1, G3) and than a frozen environment (G2, G4), the direction reported by Tsuda and Kawata. How each possible outcome would be interpreted was fixed in advance (Materials and methods).

The results did not support this prediction (Table 1, Fig 7). Under cue-free fluctuation, genome size and regulatory strength showed no detectable dependence on grain. At the coarsest grain, populations carried 19.2 genes and had a regulatory strength of 2.0, against 19.5 and 2.3 at the finest (G1 and G3, inconclusive; $p = 0.80$ and 0.60), and a secondary analysis found no trend across the four grains (Spearman correlation with log dwell time 0.06 for genes and 0.01 for strength). Coarse-grained fluctuation reduced gene number and regulatory strength relative to the frozen environment by about as much as fine-grained fluctuation did (25.6 genes and a strength of 8.4 in frozen populations; G2 and G4 refuted, Holm-adjusted $p = 4 \times 10^{-4}$ and 8×10^{-5}). Coarser grain did improve the match to the target, lowering metabolic error from 0.062 to 0.038, but genome size and regulation stayed where they were. This is the pattern expected of adaptive tracking, in which populations re-adapt genetically to each environment when it lasts many generations [9, 27]. The finest-grained cell, with new seeds, reproduced P3 and the reduction in gene number without a cue. By the interpretation fixed in advance, grain does not reconcile our results with those of Tsuda and Kawata.

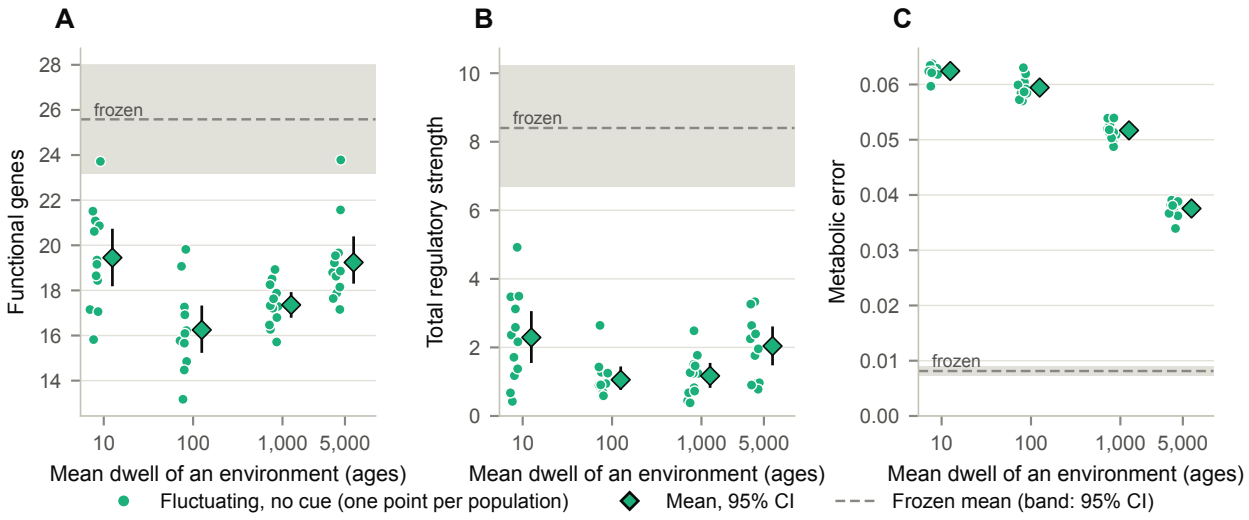


Fig 7. Environmental grain. Cue-free fluctuating populations at four switching probabilities, plotted against the mean time an environment persists ($1/q$ ages; 20 ages make one generation), one point per population, with means and bootstrap 95% confidence intervals. The dashed line and grey band show the mean and 95% confidence interval of the frozen arm. (A) Functional genes. (B) Total regulatory strength. (C) Metabolic error, the distance between phenotype and target.

Discussion

In R-Aevol, environmental fluctuation limited the genome-encoded regulatory network unless organisms could identify the current environment. In a frozen environment the network grew over tens of thousands of generations. Without a cue, or with a yoked cue that supplied the same signal proteins at the same rates, it hardly grew and reached a fraction of the frozen strength. With an informative cue it grew at least as strong as in a frozen environment, and the cue itself was wired into it, so that total regulatory strength rose. The limit depends on the current environment. Frozen networks moved into cue-free fluctuation lost most of their strength, half of the loss within 500 generations, and fluctuating networks moved into a frozen environment grew. The speed of the loss suggests that selection removed the network once the environment fluctuated without a usable cue. Slow decay would have taken longer, but we did not test this directly. The first pre-registered tests concerned total strength. They were supported across switching

rates, their contrasts held at a second population size, and the effect without a cue held across a 500-fold range of environmental grain, in tests whose predictions and analyses were fixed before the data were generated. The separation of signal-protein links from the genome-encoded network, found in an exploratory analysis and confirmed in a second pre-registered experiment, changes the interpretation of the first prediction. Most of the rise in total regulatory strength under fluctuation with a usable cue was the cue's own links, although the genome-encoded network also became somewhat stronger.

The phenotypic side of this result is well known. Reliable cues favour plastic responses, and in their absence generalist phenotypes or bet-hedging are favoured [6–9]. This has recently been shown for the evolution of plasticity in a small gene regulatory network whose link strengths also depended on cue reliability. With reliable cues the links from the regulatory genes to the output became strong, and with uninformative cues the networks ignored the cue and evolved strong links within the regulatory layer, which made them evolvable [11]. In a network with a dedicated sensor gene, plasticity evolves as the sensor comes to activate some genes and repress others [28]. Our results add a comparison with a constant environment, a measure of how much of the regulation is the cue's own wiring, and genome size. Information did not make the network larger. It let the network grow under fluctuation, made it somewhat stronger than in a frozen environment with the same signals (1.5 to 1.6 times in the confirmation), and was itself wired into it. Our yoked control separated the information from the cue's physical presence with the stimulus statistics held fixed.

Two differences from Jorritsma and van den Berg stand out. Their networks became unresponsive to uninformative cues, whereas our yoked-cue populations kept strong links from the uninformative signals, which made up most of their regulatory strength. And their networks evolved strong internal links when cues carried no information, whereas ours did not build them. Their environment changed only between generations, their network had four nodes and a fixed topology, and they compared cue reliabilities under fluctuation, without a constant environment as reference, so the two results need not conflict. The difference nevertheless deserves a direct test.

We designed the grain experiment to test whether grain accounts for the difference from Tsuda and Kawata, who found that fluctuation of the optimum across generations promotes regulatory complexity [12]. It does not. In R-Aevol, cue-free fluctuation reduced genome size and regulatory strength by similar amounts whether an environment lasted half a lifetime or about 250 generations, and coarser grain improved how closely populations matched their targets without detectably changing either quantity. Under the interpretation we fixed in advance, the remaining difference between the two studies is therefore one between the models. Going beyond that interpretation, Tsuda and Kawata's own results point to the pattern of fluctuation, which we had listed as a limitation of the design when registering it. Their strong effect came from an optimum that moved to new positions in a random walk. In their cyclic regime, which alternated between two optima as ours does, complexity rose only slightly and only for large, infrequent shifts [12], so for this kind of fluctuation the two studies differ less than their headline results suggest. A design with novel optima would test this directly. The way each model maps genotype to phenotype is a further candidate. Earlier experiments in EvoSim also produced larger genomes under fluctuation (S4 Text); coarse grain did not enlarge genomes in R-Aevol, so that difference too is most likely one between the models.

The divergence between link number and link strength in R-Aevol comes largely from the signal proteins. In frozen environments, where the signals predict nothing, they carried many weak links, and under fluctuation with an informative cue, fewer and stronger ones. We did not determine why the uninformative signals of frozen populations accumulate weak links, or why the uninformative signals of yoked populations carried stronger ones. A similar divergence appeared in EvoSim, where fluctuation raised the number of connections without raising total wiring strength (S6 Text). Many evolutionary models of regulation measure complexity by counting genes or connections [12,18], and we recommend that weighted measures be reported as well, separately for the inputs of a network and for the network itself.

Limitations

Our results come from a single model. In EvoSim, the model in which this study began, organisms did not use the cue in the regime we studied, and we have not measured regulatory strength in the regime in which

they do (S4 and S6 Text). Each R-Aevol condition contained 12 populations, fewer than the 60 to 1,000 used in comparable studies [11, 12, 29]. The effects were large ($|d|$ between 1.5 and 3.9 in the evaluated cells of the sweep and between 1.2 and 3.6 in the confirmation) and remained significant after correction for multiple testing, but smaller effects could have been missed. The pre-registered measure of regulatory strength includes the influences of the signal proteins. The first decomposition that separates them was not pre-registered and rested on continuations of each run for 2,000 generations beyond the analysis window, because resuming a run in this version of R-Aevol does not reproduce it exactly. The confirmation measured the genome-encoded network directly, with new seeds and pre-registered predictions, and supported the pattern. It tested whether the network under an informative cue is at least as strong as in a frozen environment, and its greater strength is a secondary result. The switch experiment founded each population from a single genome, which removes the standing variation of the source population; the four cells share this, and its comparisons are made within the experiment. Frozen networks had not levelled off by generation 30,000. Between generations 20,001–25,000 and 25,001–30,000 the regulatory strength of the 48 cue-free frozen populations rose by a further 3.6% ($p = 0.006$), whereas that of the cue-free fluctuating populations did not change, so the differences we report are those after 30,000 generations and not at equilibrium. This version of R-Aevol records per-generation statistics only for the best individual in each population, so every R-Aevol measure describes the fittest individual of each generation, averaged over the analysis window, and not the population average. R-Aevol's fluctuating environments alternate between two fixed targets and do not include the random walk towards new optima that Tsuda and Kawata found most favourable to complexity [12], and even at the coarsest grain a switch falls within about 0.4% of lifetimes. Prediction P4 establishes the direction of the effect on genome size; its mechanism is untested. Finally, all pre-registrations were committed to a version-control repository before the corresponding runs, but these timestamps exist only locally and cannot be verified independently.

Conclusion

Environmental fluctuation did not by itself strengthen gene regulation in R-Aevol. Without information about the environment it kept the regulatory network from growing, whatever the grain of the environment, and removed networks that already existed. With an informative cue the network grew at least as strong as in a frozen environment and the cue was wired into it, so that total regulatory strength rose. A pre-registered confirmation supported this pattern. Because link counts and link strength, and the links of a network's inputs and of the network itself, can change independently, conclusions about regulatory complexity may depend on which is measured.

Materials and methods

R-Aevol

We used the `aevol-7` branch of Aevol (commit `33277374`, https://gitlab.inria.fr/aevol/aevol) compiled with regulation enabled (`-Dwith-raevol=ON`) and with per-generation statistics (`-Dwithout-stats=OFF`). Building on Apple Silicon required correcting a hard-coded path to the OpenMP library (S2 Text). The parameters (Table 3) are taken from the example configurations distributed with the platform (`basic_regul`, `dynamic_env` and `complex_env`). The mutation and rearrangement rates are those of `complex_env`, ten times those of `dynamic_env`, and the binding matrix is the file supplied with `dynamic_env`, used unchanged in every run.

Environments. Each environment is a target function made of three Gaussians. Target B is target A reflected about the midpoint of the phenotypic axis, and the two have the same geometric area, 0.152468. Fluctuating arms switch between A and B. Frozen arms use two copies of the same target (A for odd seeds, B for even seeds), so that the switching mechanism runs in the same way but the organism never experiences a change. The environment is a Markov chain over ages that continues across generations. At each of the 20 ages it moves, with probability q, to a different environment chosen uniformly among the

others. We checked in the source code that this is the switching carried out by the platform's SIMD engine (S2 Text).

Signals and the yoked control. Two signal proteins, taken unchanged from the `complex_env` example, are present at concentration 0.9 during the ages in which their environment is active and absent otherwise. The informative arm has two environments, (A, s1) and (B, s2). The yoked arm has four, (A, s1), (A, s2), (B, s1) and (B, s2). The no-cue arm has no signals. In the two-environment arm every switch changes the target and the signal together, so each changes at rate q per age. In the four-environment arm a switch changes the target with probability 2/3 and the signal with probability 2/3, so with a switching probability of $1.5q$ each again changes at rate q per age.

Measures. R-Aevol records statistics of the best individual in each generation. Genome size is the number of functional genes. Total regulatory strength is $S = n_{\text{link}}\,\bar{a}$, where n_{link} is the number of activating and repressing protein–RNA influences (a protein that activates and also represses an RNA counts twice) and $\bar{a}$ is their mean strength (the sum of the activation and repression factors divided by their number). $S = 0$ when there are no links. R-Aevol treats the signal proteins like other proteins when it computes binding, so in arms with a cue n_{link} and S include influences of the signal proteins, which the statistics do not separate. Each run is summarised by its mean over generations 25,001 to 30,000.

Runs. The exploratory experiment had six arms with 12 seeds each at $q = 0.10$ and a population of 1024 (72 runs). The pre-registered sweep comprised fluctuating informative, yoked and no-cue arms and frozen informative arms at $q = 0.05$, 0.10 and 0.20, together with frozen no-cue and frozen yoked arms at $q = 0.10$ (the frozen no-cue arm does not depend on q), each with 12 new seeds and a population of 1024 (168 runs). The fluctuating informative, yoked and no-cue arms and the frozen informative arm were repeated at a population of 512 (48 runs). The grain experiment has cue-free fluctuating arms at $q = 0.1$, 0.01, 0.001 and 0.0002 and a frozen arm, with 12 new seeds each (60 runs). All runs reached 30,000 generations.

Signal-protein links (exploratory). To separate the influences of the signal proteins from those of genome-encoded proteins, we compiled a copy of the platform whose statistics also report the number of positive activation and repression factors exerted by signal proteins and their sum (S2 Text). Resuming a run from a saved population with this version did not reproduce the original run exactly. A run resumed at generation 20,000 differed from the original from the first resumed generation on. We therefore continued each run from its saved population at generation 30,000 to generation 32,000 in the same environment and summarised it over those 2,000 generations. We did this for all 168 primary runs of the pre-registered sweep, the 36 runs with signals at a population of 512 and the 48 exploratory runs with signals (252 runs). The genome-encoded part of the regulatory strength is S minus the sum of the signal proteins' factors, and likewise for link counts. Contrasts were tested with the same permutation test, without correction and without verdicts.

Confirmation. The second pre-registered experiment used the instrumented version to create and run populations, so the signal proteins' influences were recorded from generation 0. It had fluctuating informative, yoked and no-cue arms and frozen informative and yoked arms at $q = 0.10$ and 0.20, and a frozen no-cue arm, each with 12 new seeds and a population of 1024 (132 runs). Every other setting was that of the pre-registered sweep. All runs reached 30,000 generations and were summarised over generations 25,001 to 30,000. The genome-encoded strength of the no-cue arms is their total strength, since they have no signals.

Trajectories (exploratory). For every run of the four experiments, the statistics of each generation were averaged over consecutive blocks of 500 generations. Regulator proteins are genome-encoded proteins with at least one regulatory link (the platform's `nb_TF`). Changes within runs were tested with paired sign-flip permutation tests (100,000 permutations).

Environment switch. The third pre-registered experiment started from the cue-free cells of the confirmation at $q = 0.10$. The best individual of each run at generation 30,000 was extracted from the saved population (`aevol_post_extract`) and founded a clonal population of 1024 (`aevol_create -C`), which evolved for 10,000 generations with the instrumented version in one of four cells of 12 runs. Frozen populations were kept frozen or moved to cue-free fluctuation at $q = 0.10$, and fluctuating populations were kept fluctuating or moved to a frozen environment (target A for odd and B for even replicates, as in the other frozen arms). Every other setting was that of the sweep. In a pilot, a frozen genome refounded in its own environment reproduced the statistics of its source at generation 30,000 exactly. Regulatory strength at the start and the end is the mean over generations 1 to 500 and 8,001 to 10,000.

Table 3. R-Aevol parameters.

Parameter	Value
Population	1024 (32×32) or 512 (32×16)
Selection	fitness-proportionate, pressure 750
Point mutation, small insertion, small deletion	5×10^{-5} per base per generation
Duplication, deletion, translocation, inversion	5×10^{-5} per base per generation
Maximum indel size	6
Initial genome	5,000 bases, one functional gene
Hill function	$n = 4$, $\theta = 0.5$
Degradation	rate 1, 10 steps per age
Lifetime	20 ages, fitness evaluated at all 20
Binding matrix	fixed file from `examples/dynamic_env`, all runs
Target sampling	300 points
Signal concentration	0.9
Generations	30,000

Statistics

The unit of analysis is a population. Each run contributes one value per measure, so generations within a run are not treated as independent. Differences between arms were tested with two-sided permutation tests on the difference of means (100,000 permutations). Rank-based permutation tests of the pre-registered contrasts led to the same verdicts (S2 Text). Multiple predictions were corrected with the Holm procedure. Effect sizes are given as Cohen's d, and intervals in the figures are bootstrap 95% confidence intervals (10,000 resamples).

Pre-registration and deviations

Predictions P1 to P4, their measures, tests and decision rules, and the analysis code were committed to version control (commit `e6173aa`) before the pre-registered sweep was started. The analysis was run once, on the complete data, with the committed code, whose checksum was verified beforehand. The only deviation was operational. The idle-sleep setting of the host machine suspended the runs overnight, which lengthened the wall time but cannot have affected the results, because a suspended process resumes from the state in which it stopped. A pre-registered confirmation in EvoSim could not be run, because no candidate regime met its viability condition (S6 Text). The design, predictions and analysis code of the grain experiment were committed in the same way (commit `aafb136`) before its runs began, and its analysis was run once, automatically, when the last run finished. Its interpretation was fixed at the same time. If coarse grain gave more genes (or stronger regulation) than fine grain and than the frozen environment, grain would reconcile the two studies. If it did so only relative to fine grain, Tsuda and Kawata's gain would probably need novel optima, which our design cannot produce. If it did not do so relative to fine grain, grain would not be the explanation and the difference would be reported as one between the models. The confirmation (C1 to C4) was pre-registered in the same way (commit `e15104a`),

after the exploratory decomposition and before any of its runs. Its analysis script, tested beforehand on synthetic data, fixed the tests, the decision rules and what each outcome would mean for the paper, including a change of title if the informative cue did not maintain the network. The script was run once, after its checksum was verified, when the last run finished. The environment-switch experiment was pre-registered in the same way. Its design and the interpretation of each outcome were committed before the runs (commit `5bd4fee`), and its analysis script, tested on synthetic data, was committed after the launch but before any output was inspected (commit `baeddca`); it ran once when the last run finished.

Software and availability

The R-Aevol runs were made on a single 12-core machine. EvoSim (S1 and S6 Text) is written in Julia and uses only the standard library for simulation. The configurations of the two models, the launch scripts, the pre-registered analysis scripts, the instrumented statistics and continuation scripts, the history of all analyses and the per-run statistics are available at https://github.com/iosmanRJ/fluctuation-limits-regulation and archived on Zenodo (https://doi.org/10.5281/zenodo.24054019; version 1.0, https://doi.org/10.5281/zenodo.24054065, was used for this paper).

Supporting information

S1 Text. ODD description of EvoSim. Complete model description following the ODD protocol, with every parameter value.

S2 Text. R-Aevol build, configuration and the yoked-control derivation. Build options and the changes needed on Apple Silicon, the parameter-file template, targets and signals, the environment-switching code, the derivation of matched rates of target and signal change, the definition of the measures, seeds, the pre-registration record, and the result of every pre-registered test with rank-based p values.

S3 Table. Results for every arm of every experiment. Means and standard deviations for all 708 runs and the 252 continuations of the signal-link decomposition. Per-run values are in the repository.

S4 Text. Earlier EvoSim results and the audit trail. Status of earlier claims after this study, the record of pre-registered predictions, and the measurement errors found during the project with the checks that exposed them.

S5 Table. Prior work and how this study relates to it.

S6 Text. EvoSim: connection count and wiring strength under fluctuation (exploratory). The EvoSim experiment, its methods, the population-size confound and the pre-registered confirmation that could not be run.

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